Research Methodology in Computer Science

UNIT I WORKSHEET

Introduction to Research Problem



Course: CSEG3060 | Unit: I | Lecture Allocation: 9 Hours
Designed as an active-learning revision and practice worksheet.

Name: Krish Aggarwal **Roll No.:** 590014301

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How to use this worksheet

This worksheet is meant to move you from *"I know the definitions"* to *"I can formulate and defend a research problem."* Work through the activities in order. Use your own Computer Science topic whenever a question asks you to design or formulate something.

Unit I Coverage Checklist

- ☑ Meaning and significance of research
- ☑ Objectives of research
- ☑ Key elements of research
- ☑ Sources of research problems
- ☑ Identification of research problems
- ☑ Good research problem criteria
- ☑ Common selection errors
- ☑ Avoiding selection errors
- ☑ Scope and boundaries
- ☑ Research objectives
- ☑ SMART objectives
- ☑ Research approaches
- ☑ Theoretical vs. empirical research
- ☑ Data types and sources
- ☑ Data collection tools
- ☑ Data collection guidelines
- ☑ Data analysis
- ☑ Interpretation of results
- ☑ Research instrumentation
- ☑ Calibration and repeatability
- ☑ Integrated case-based application

1 Part A — Research: What, Why and How?

A1. Research in one sentence

Research is a **systematic and scientific investigation** used to discover new facts, verify existing knowledge, analyze relationships, explain phenomena, and answer questions using evidence.
Key elements from class:

Problem Identification → Data Collection & Analysis → Hypothesis Formation → Result Interpretation

Activity 1: Build the research chain

Put the following in a sensible order:

Result Interpretation	Problem Identification	Data Collection/Analysis	Hypothesis Formation
4	1	2	3

Why is this order useful in a CS research project? **[2 marks]**

It follows the logic of the scientific method: first define a clear problem, then collect and analyze data, form/test a hypothesis, and finally interpret results. In CS it prevents building or testing without a goal, so experiments (e.g., SUMO runs) stay focused, measurable and reproducible.

Activity 2: Why does research matter?

Give one CS example for each:

Technology/knowledge advancement	Deep learning (CNNs, transformers) advanced image recognition and language understanding.
Industrial innovation	Route-optimization ML in ride-hailing/maps apps cuts delivery time and fuel cost.
Academic growth	Papers on new algorithms (e.g., federated learning) in IEEE/ACM venues build academic knowledge.
Policy or societal decision-making	AI-based traffic analysis helps city authorities decide on signal policy, congestion and emission control.
Computer Science example (AI/algorithms/HCI/etc.)	Reinforcement learning for adaptive traffic-signal control in smart cities.

Activity 3: Research objectives

Complete the table.

Objective of research	Your CS example
Find answers using scientific methods	Test with controlled SUMO experiments whether a DQN signal controller lowers vehicle waiting time.
Explain phenomena / establish relationships	Analyze how traffic arrival rate affects queue length and delay at an intersection.
Test or validate an existing theory	Validate the theory that adaptive control outperforms fixed-time control in peak traffic.

Think like a researcher

A student says: *"I want to build an AI app because AI is trending."* Is this already a research problem? Explain what is missing.

No. "AI is trending" is only a motivation or topic, not a research problem. Missing: a specific problem/pain-point, a research gap, a clear research question, objectives, scope, data/evaluation method and measurable outcomes.

2 Part B — Types and Approaches of Research

Research approaches covered in Unit I

Exploratory
discover patterns/ideas

Descriptive
describe facts/characteristics

Analytical
analyze information

Theoretical
models, logic, proofs

Empirical
observation/experiment

The class material also introduced **fundamental** and **applied** research as broad research types. Choose the approach according to the research question and the kind of evidence required.

Activity 4: Match the approach

Write E (Exploratory), D (Descriptive), A (Analytical), T (Theoretical), or EM (Empirical).

1. Investigating an emerging problem where very little is known: E
2. Measuring average response time of five search algorithms: EM
3. Developing a formal model for a new computational system: T
4. Summarizing user characteristics from a usability survey: D
5. Comparing existing datasets and identifying relationships among variables: A

Activity 5: Theoretical or empirical?

For each scenario, decide the dominant approach and justify it in one sentence.

Scenario	Type	Why?
Prove a property of an algorithm mathematically	Theoretical	Proof relies on mathematical logic and reasoning, not on running experiments.
Benchmark an intrusion-detection model on a dataset	Empirical	Performance is measured by running the model on data and recording metrics.
Develop a formal model of a distributed protocol	Theoretical	It builds an abstract, formal model of the protocol using logic/specification.
Interview developers to discover pain-points in an emerging tool	Empirical (exploratory)	Interviews gather real observations to explore an emerging area where little is known.

3 Part C — Finding and Formulating a Research Problem

C1. Where do research problems come from?

Important sources highlighted in class include:

- Literature reviews and research gaps.
- Industrial problems and real-world needs.
- Emerging trends and technologies.
- Academic coursework and past projects.
- Surveys of recent publications.
- Patent and research databases.
- Industry white papers.
- Competitive research proposals.

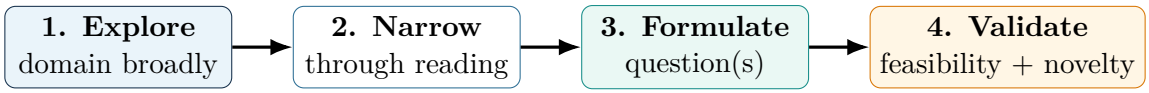
Activity 6: Research-problem detective

Pick a CS domain: AI / Smart-city traffic management

List **four possible sources** from which a research problem in this domain could emerge:

1. Literature review / research gaps: recent papers on reinforcement learning for traffic-signal control show gaps such as poor generalization to changing traffic.
2. Industrial problems and real-world needs: city traffic departments face congestion, long waiting times and fuel wastage at fixed-time signals.
3. Emerging trends and technologies: IoT sensors, connected vehicles and edge AI now make real-time adaptive signal control possible.
4. Patent/research databases and industry white papers: existing adaptive signal systems reveal unsolved limitations that can be studied.

C2. Identification workflow



Activity 7: Apply the four-step workflow

Topic/domain: AI-based adaptive traffic-signal control

1. **Explore broadly:** What is the larger area?
   Smart-city traffic management: using AI and sensor data to make urban road networks more efficient and less congested.
2. **Narrow through reading:** What specific sub-area or pain-point interests you?
   Fixed-time signals cannot adapt to changing traffic, so vehicles wait long at busy junctions. The sub-area is reinforcement-learning-based adaptive signal control.
3. **Formulate question(s):** Write one research question.
   Can a DQN-based adaptive signal controller reduce average vehicle waiting time by at least 15% compared with fixed-time control at a 4-way intersection?
4. **Validate:** What would you check for feasibility and novelty?
   Feasibility: free tools (SUMO, Python), open traffic data, laptop-scale compute, 12 weeks. Novelty: search IEEE/Scopus for existing DQN-signal studies to confirm a gap remains.

Quick Pair Challenge: Problem or Topic?

Classify each statement as **Topic**, **Weak Problem**, or **Research Problem**.

1. "Artificial Intelligence in Smart Cities." Topic
2. "AI is interesting." Weak Problem
3. "How can reinforcement learning optimize traffic signals in smart cities?" Research Problem
4. "Cybersecurity." Topic
5. Your own statement: DQN signal control to cut junction delay

C3. Characteristics of a good research problem

A strong problem should be **clear and concise, researchable and feasible, relevant and original**, and should offer scope for **data collection and analysis**. The syllabus also emphasizes essential attributes and avoiding errors in selection. **[5 marks]**

Criterion	Evidence that YOUR problem satisfies it
Clear and concise	The problem fits one clear sentence: reduce average waiting time at a 4-way intersection using a DQN signal controller.
Researchable	It can be investigated with simulation experiments and measurable outcomes (waiting time, queue length, throughput).
Feasible	Uses free tools (SUMO, Python, PyTorch), runs on a laptop and fits within a 12-week semester.
Relevant	Congestion causes delay, fuel waste and emissions; adaptive signals address a real smart-city need.
Original / non-trivial	Goes beyond "use AI": compares DQN with fixed-time and actuated control under peak and off-peak demand.
Allows data collection and analysis	SUMO logs waiting time, queue length and throughput, which can be statistically analyzed over repeated runs.

4 Part D — What Can Go Wrong? Scope and Boundaries

Common errors in selecting a research problem

1. Ambiguity / vagueness
The problem does not clearly state what is being investigated.
2. Too broad
The study becomes unrealistic or unmanageable.
3. Too narrow
There may be insufficient room for meaningful investigation.
4. Lack of relevance
The problem has weak academic, practical, or societal value.
5. Lack of feasibility
Resources, data, skills, time, or infrastructure are unavailable.
6. Ignoring prior art
Existing research may already answer the question.

Activity 8: Error clinic

For each weak problem, identify the main error and rewrite it.

1. "Improve AI everywhere."
   Error: Too broad / vague (no clear focus)
   Improved version: Evaluate whether transfer learning improves the accuracy of a CNN for detecting road damage on a public road-image dataset.

2. "Study all cybersecurity threats affecting every organization worldwide."
   Error: Too broad (unmanageable scope)
   Improved version: Compare three machine-learning classifiers for detecting phishing emails on a public labelled dataset within six months.

3. "Build a system that detects every possible disease with 100% accuracy in two days."
   Error: Lack of feasibility (unrealistic claim)
   Improved version: Develop and evaluate a CNN to detect pneumonia from chest X-rays on a public dataset, targeting at least 90% accuracy in 8 weeks.

How to avoid selection errors

- Perform a literature review.
- Consult experts or guides.
- Conduct pilot studies where appropriate.
- Align the problem with objectives and available resources.

D2. Scope = the boundary of your research

Scope defines the **limits and boundaries** of the study and prevents deviation from the objectives. Typical boundaries include:

Time frame | **Geographic** location | **Target population** or dataset | **Tools /** technologies

Activity 9: Draw the boundary

Research problem: *"How can machine learning improve network intrusion detection?"*

Fill in realistic boundaries:

Time frame	12 weeks (one semester): data preparation, model training and evaluation.
Geographic / deployment context	One simulated campus/enterprise network; offline evaluation on a lab machine, no live deployment.
Target dataset / population	Public labelled network-traffic dataset such as NSL-KDD or CICIDS2017.
Tools / algorithms	Python, scikit-learn; Random Forest, SVM and a small neural network; Wireshark for inspecting traffic.
What is explicitly OUT of scope?	Real-time deployment, encrypted-traffic analysis, IoT/cloud networks, hardware IDS and zero-day discovery.

Activity 10: Scope test

A student's project is titled *"AI for Smart Cities."* Give three reasons why this scope is likely too broad, then convert it into a narrower study.

Why "AI for Smart Cities" is too broad:
1. "AI" covers many techniques (ML, NLP, vision) and "smart cities" covers many sectors (traffic, waste, energy, water, safety), so there is no single clear problem.
2. No specific data, population or time frame is defined, so it cannot be completed with limited time and resources.
3. There is no measurable objective, so results cannot be evaluated or compared.

Narrower study: Using reinforcement learning (DQN) to reduce average vehicle waiting time at a single 4-way intersection in a SUMO simulation over 12 weeks.

5 Part E — Research Objectives and SMART Thinking

E1. Why objectives matter

Research objectives:

- set the direction and purpose of research;
- help define scope and methodology; and
- provide focus and coherence.

A good objective is not just an activity such as "study AI." It states **what will be achieved or evaluated**.



Activity 11: SMART objective laboratory

Research topic/problem: AI-based adaptive traffic-signal control using reinforcement learning (DQN)

Write one objective, then audit it.

Specific	What exactly will be done? Train a DQN controller in SUMO to control a 4-way intersection and compare it with fixed-time control.
Measurable	What variable, metric, or evidence will show achievement? Average vehicle waiting time (s), queue length and throughput; target: at least 15% reduction in waiting time.
Achievable	Why is it realistic with available resources? Free tools (SUMO, Python, PyTorch); a single intersection trains on a laptop in modest time.
Relevant	How does it address the research problem? It directly tests whether AI adaptive control reduces congestion at signals, the core research problem.
Time-bound	By when / over what period? Within 12 weeks: 4 weeks simulation setup, 6 weeks training, 2 weeks evaluation.

Final SMART objective:

Within 12 weeks, train and evaluate a DQN-based traffic-signal controller in SUMO for a single 4-way intersection and achieve at least a 15% reduction in average vehicle waiting time compared with fixed-time control, averaged over 20 simulation runs.

Peer Review — 60-second test

Exchange your objective with a partner. Your partner must answer:

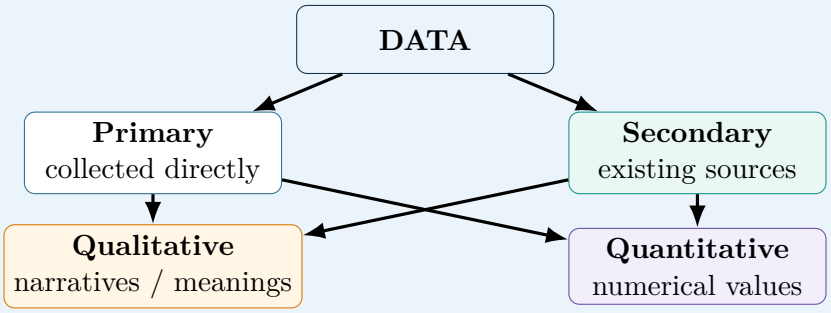
1. Can I identify exactly what is being investigated?
2. Can I identify how success will be measured?
3. Is the target realistic?
4. Does it directly connect to the problem?
5. Is a time boundary visible?

Partner's one improvement suggestion:

Also state the traffic demand levels (peak / off-peak) and the statistical test used to confirm the 15% reduction.

6 Part F — Data Fundamentals

F1. Two useful ways to classify data



Examples discussed in class include primary interviews, sensor readings, and usage logs; secondary GitHub/Kaggle datasets, IEEE articles, and government data.

Activity 12: Classify the data

Mark each as Primary/Secondary and Qualitative/Quantitative where applicable.

Data source	Primary/Secondary	Qualitative/Quantitative
Interview transcripts	Primary	Qualitative
Sensor temperature readings	Primary	Quantitative
Kaggle dataset downloaded for analysis	Secondary	Quantitative
System CPU utilization logs collected during your experiment	Primary	Quantitative
User comments about usability	Primary	Qualitative
IEEE article used as an existing source	Secondary	Qualitative (text)

F2. Choosing data

For your own research problem:

1. What data do you need?
   Traffic data at a 4-way intersection: vehicle arrival rates, waiting times, queue lengths, signal phases and throughput.
2. Why is this data appropriate?
   These variables directly measure congestion and signal performance, so they show whether the AI controller improves traffic flow.
3. Primary, secondary, or a combination?
   A combination: primary data from my own SUMO simulation runs, plus secondary public traffic-count data to set realistic demand.
4. Qualitative, quantitative, or a combination?
   Mainly quantitative (numerical metrics), with a small qualitative part from literature review.

7 Part G — Data Collection Tools and Guidelines

G1. Match the tool to the evidence

The class material highlighted:

- **Questionnaires / online forms** — structured responses from participants.
- **Observation logs / experiments** — record observed behavior or experimental events.
- **System monitoring tools** — capture system/network performance.
- **Web scraping** — collect information from web sources using appropriate tools.

Examples shown in the class material include Google Forms, Typeform, Wireshark, PerfMon, BeautifulSoup, and Selenium.

Activity 13: Tool selection game

You need to collect the following evidence. Select the most suitable tool and justify.

Evidence required	Tool / method	Why?
Students' satisfaction with an app	Questionnaire (Google Forms / Typeform)	Gives structured, easily quantified responses from many students.
Network packet behavior during an experiment	Wireshark	Captures and inspects packets flowing through the network during the experiment.
CPU/memory usage of a program	PerfMon (system-monitoring tool)	Records CPU and memory usage of the program over time.
Behavior of users interacting with a prototype	Observation log / experiment log	Records how users actually behave while interacting with the prototype.
Public web data needed for a study	Web scraping (BeautifulSoup / Selenium)	Automatically collects large amounts of public web data.

Activity 14: Design a mini data-collection instrument

Choose one: ☐questionnaire ☐observation log ☒experiment log ☐system-monitoring plan.

Research problem: Reducing vehicle waiting time at a 4-way intersection using DQN-based signal control

What will be measured/recorded?	Per episode: average waiting time (s), queue length (vehicles), throughput (veh/h) and signal-phase changes.
Who/what is the source?	SUMO simulation of a 4-way intersection (TraCI output logs), with demand set from public traffic-count data.
When/how often?	Logged every simulation step (1 s), summarized per episode; 20 runs per controller for peak and off-peak demand.
What tool will be used?	SUMO + TraCI (Python) writing CSV logs; analysis with pandas.
How will consistency be maintained?	Fixed random seeds, identical demand files and parameters for all controllers, same warm-up period, version-controlled scripts.

8 Part H — From Data to Meaning: Analysis and Interpretation

H1. Do not confuse these stages

Collect raw evidence → **Analyze** organize/compare → **Interpret** explain meaning → **Conclude** answer problem

Analysis asks: "What patterns, differences, relationships, or measurements are in the data?"
Interpretation asks: "What do those findings mean for my research problem and objectives?"

Activity 15: Read the result, then interpret it

Suppose an experiment gives:

Model	Accuracy	Inference time (ms)
A	92%	20
B	95%	70
C	94%	25

1. Which model has the highest accuracy?
   Model B (95%).
2. Which model is fastest?
   Model A (20 ms).
3. If the system is real-time and accuracy must be at least 93%, which model would you investigate first? Why?
   Model C: it meets the 93% accuracy requirement (94%) and is fast (25 ms). B is too slow (70 ms) for real-time use and A is below 93%.
4. Is "B is best" a valid conclusion without considering the research objective? Explain.
   No. "Best" depends on the objective and constraints: B has the highest accuracy but the slowest inference, so it may fail a real-time requirement.

Activity 16: Interpretation vs. description

Complete each sentence with either **description** or **interpretation**.

1. "Model B achieved 95% accuracy." description
2. "The higher accuracy may be useful when accuracy is the primary objective, but its latency may limit real-time deployment." interpretation
3. "The system processed 10,000 requests." description
4. "The observed throughput suggests the system can support the target workload." interpretation

Researcher's caution

A graph can show a pattern, but a pattern is not automatically an explanation. Write one question you would ask before claiming *why* a result occurred.
Did another factor (data change, hardware load, random seed or noise) cause this pattern, and does it repeat when I rerun the experiment?

9 Part I — Research Instrumentation

I1. Instrumentation in computing research

Research instrumentation refers to the **tools, hardware, and software used for data collection and analysis**. Class examples include:

- sensors;
- simulators;
- benchmarking suites;
- custom tools for robotics, networks, and AI.

Two important quality ideas emphasized in the class material are **calibration** and **repeatability**.

Activity 17: Instrument selection matrix

For your chosen research problem, identify one instrument at each level.

Level	Possible instrument	What evidence will it produce?
Hardware / sensor	Roadside camera / inductive loop vehicle detector	Vehicle counts, speeds and queue presence at the intersection.
Software / simulator	SUMO traffic simulator with TraCI (Python API)	Simulated vehicle trajectories, waiting times and signal-phase logs.
Benchmark / measurement tool	Metric scripts: waiting time, queue length, throughput; fixed-time and actuated baselines	Comparable performance numbers across controllers.
Custom tool (if needed)	Python/Gym environment wrapping SUMO for DQN training and logging	Rewards, episode returns, training curves and CSV logs.

Activity 18: Calibration and repeatability

Why should a sensor, measurement setup, benchmark, or custom instrument be calibrated?

Calibration compares an instrument with a known standard so readings are accurate and unbiased, e.g., checking that detector or simulated vehicle counts match actual counts and that the SUMO demand matches observed traffic.

What could happen if the same experiment cannot be repeated consistently?

Results become unreliable and others cannot verify them, so conclusions may be due to chance or setup errors. The findings lose credibility and cannot be generalized or reproduced.

10 Part J — The Research Problem Builder

Your 10-minute research design

Choose a real Computer Science problem you care about. Complete every box.

Step 1 — Domain

AI / Smart-city traffic management (reinforcement learning for adaptive traffic-signal control)

Step 2 — Real-world pain-point

Fixed-time signals cause long waiting times, congestion, fuel waste and emissions at busy junctions.

Step 3 — Research problem statement

Fixed-time signals cannot adapt to real-time traffic; it is unclear how much a DQN controller can improve this.

Step 4 — Research question

Can a DQN-based signal controller cut average waiting time by 15% or more versus fixed-time control at a 4-way junction?

Step 5 — Why is it worth investigating?

It reduces delay, fuel use and emissions; adaptive control is a practical, low-cost smart-city improvement.

Step 6 — Research approach

☐Exploratory ☐Descriptive ☐Analytical ☐Theoretical ☒Empirical ☐Fundamental ☒Applied

Step 7 — Scope boundaries

Time: 12 weeks Dataset/population: SUMO 4-way junction; 20 runs/controller

Tools/technology: Python, SUMO, TraCI, PyTorch Out of scope: Real hardware; pedestrians

Step 8 — Three SMART objectives

O1: Build a SUMO model of a 4-way junction with peak/off-peak demand and a fixed-time baseline within 4 weeks.

O2: Train a DQN controller within 8 weeks to cut average waiting time by at least 15% over 20 runs vs the baseline.

O3: By week 12, compare DQN, fixed-time and actuated control on waiting time, queue length and throughput (mean +/- SD).

Step 9 — Data plan

Data type(s): Quantitative (primary + secondary) Source(s): SUMO runs; public traffic counts

Collection tool(s): SUMO + TraCI (Python), pandas Instrumentation: SUMO simulator, Python logger

Step 10 — Analysis and interpretation

What will you measure/analyze? How will the findings answer the research question?

Measure average waiting time, queue length and throughput per episode for each controller and compare means over 20 runs (SD / t-test). Interpret whether the 15% target is met, and when it fails (peak vs off-peak), to answer the research question.

11 Part K — Case Study: AI in Smart Cities

Case Study — Refine the vague idea

A student wants to research:

"Artificial Intelligence in Smart Cities."

Your task is to transform this broad idea into a defensible Unit I research design.

K1. Narrow the problem

Broad area: **AI + Smart Cities**

Choose one specific application: ☒traffic ☐waste ☐energy ☐water ☐safety ☐other: _____

Specific pain-point: Fixed-time signals ignore real-time demand, so vehicles wait long at 4-way junctions, especially at peak hours.

Refined research problem: How effectively can a DQN-based reinforcement-learning controller reduce average vehicle waiting time at a single 4-way intersection compared with fixed-time control, in a SUMO simulation over 12 weeks?

K2. Define the approach

Dominant approach: Empirical (applied)

Why is it theoretical, empirical, exploratory, descriptive, analytical, or a combination?

Empirical: it is based on simulation experiments and measured data (waiting time, queue length). It is also applied (solves a practical problem) and partly analytical, as different controllers are compared.

K3. Three SMART objectives

1. Build a SUMO model of one 4-way intersection with peak/off-peak demand and a fixed-time baseline within 4 weeks.
2. Train a DQN controller within 8 weeks to cut average waiting time by at least 15% versus the baseline over 20 runs.
3. By week 12, evaluate DQN against fixed-time and actuated control on waiting time, queue length and throughput (mean +/- SD).

K4. Boundaries

Timeframe	12 weeks (one semester)
Target dataset / population	Simulated traffic of one 4-way intersection (peak/off-peak flows) set from public traffic-count data; 20 runs per controller.
Geographic/deployment context	Single urban 4-way intersection (simulated); no live deployment.
Tools/technologies	Python, SUMO, TraCI, PyTorch / Stable-Baselines3, pandas.
Out of scope	Multi-intersection networks, pedestrians and emergency vehicles, real hardware deployment, V2X communication.

12 Part L — Case-Based Thinking

Case 1: "Build a better chatbot"

A student proposes: *"I will make a chatbot that is better than ChatGPT."*
Identify at least **four** problems with this research proposal at the Unit I level.

1. Too broad / vague: "better" is not defined (faster, more accurate, safer?).
2. No measurable objective or evaluation metric, so success cannot be tested.
3. Not feasible: ChatGPT needs huge data, compute and expert teams that a student lacks.
4. Ignores prior art; no research gap or question, it is a product idea, not a research problem.

Rewrite it as a focused research problem:
Evaluate whether fine-tuning a small open-source language model on a college FAQ dataset improves answer accuracy over its base version, measured on a held-out test set within 8 weeks.

Case 2: Sensor-based smart agriculture

A team wants to predict irrigation needs using soil-moisture sensors.
Identify:

1. One possible research problem. Predicting irrigation needs from soil-moisture readings to reduce water use.
2. Two data sources. Soil-moisture sensor readings; public weather/rainfall data.
3. One primary data source. Soil-moisture readings collected from the field sensors.
4. One secondary data source. Public weather records (e.g., government data) or a Kaggle crop dataset.
5. One data-collection tool. ESP32 data logger sending readings to a database.
6. One research instrument. Capacitive soil-moisture sensor (calibrated against soil samples).
7. One measurable objective. Predict irrigation need with at least 90% accuracy within 10 weeks on one plot.
8. One scope boundary. Limited to one crop type and one field plot in one growing season.

Case 3: The "perfect accuracy" trap

A researcher claims: *"My model is successful because it gives 99.9% accuracy."*
Write three questions you would ask before accepting the claim:

1. Was 99.9% measured on unseen test data, or on training data (overfitting / data leakage)?
2. Is the data balanced? What are precision, recall and F1, since accuracy alone can mislead?
3. Can others repeat the result, and how does it compare with baseline models?

13 Part M — Unit I Rapid-Fire Quiz

Choose the best answer.

1. A research problem should primarily be:

   (a) fashionable
   (c) extremely broad
   (b) clear and researchable
   (d) impossible to test

2. Which is a source of research problems?

   (a) literature gaps
   (c) avoiding prior work
   (b) random guessing only
   (d) unrelated facts

3. "Measure the average latency of Algorithm X on Dataset Y within four weeks" is closest to:

   (a) vague topic
   (c) literature review
   (b) SMART objective
   (d) research instrument

4. Existing data from Kaggle is generally:

   (a) primary
   (c) always qualitative
   (b) secondary
   (d) always experimental

5. "User comments about usability" are primarily:

   (a) qualitative
   (c) hardware
   (b) quantitative
   (d) benchmark

6. Which is most directly related to system monitoring?

   (a) PerfMon
   (c) patent claim
   (b) title page
   (d) abstract

7. A study based mainly on observation and experiments is:

   (a) empirical
   (c) descriptive writing
   (b) purely theoretical
   (d) scope definition

8. Which is NOT a good scope boundary?

   (a) dataset
   (c) tools
   (b) timeframe
   (d) "everything"

9. Interpretation primarily asks:

   (a) What does the result mean?
   (c) Who printed the report?
   (b) What font was used?
   (d) What is the page number?

10. Calibration and repeatability are especially important for:

(a) research instrumentation

(c) roll numbers

(b) research titles

(d) acknowledgements

Rapid-fire: One-word / one-line

1. Define research in your own words:
   Research is a systematic, scientific investigation that uses evidence to discover new facts, verify knowledge and answer questions.
2. One difference between a topic and a research problem:
   A topic is a broad area (e.g., "AI in smart cities"); a research problem is a specific, researchable question with a gap and a measurable scope.
3. One source of a research problem: Literature review / research gaps
4. One common selection error: Too broad (or vague) problem
5. Expand SMART: Specific, Measurable, Achievable, Relevant, Time-bound
6. One primary-data example in CS: Waiting-time logs from my own SUMO experiment
7. One secondary-data example in CS: Kaggle / GitHub dataset or an IEEE article
8. One data collection tool: Google Forms (questionnaire) / Wireshark
9. One research instrument: SUMO simulator / benchmark suite / sensor
10. Difference between analysis and interpretation:
    Analysis finds patterns and measurements in the data (what the data shows); interpretation explains what those findings mean for the research problem.

14 Part N — Connect the Dots

Match Column A with Column B.

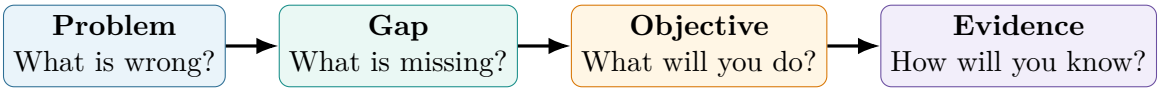
A	Concept	B	Meaning / Example
1	Exploratory	A	Collected directly
2	Primary data	B	Clear, measurable, realistic, relevant, time-bound
3	Scope	C	Discover patterns / new ideas
4	SMART	D	Limits and boundaries of a study
5	Empirical	E	Observation / experiment based
6	Instrumentation	F	Tools/hardware/software for collection and analysis
7	Secondary data	G	Existing datasets/articles/logs
8	Interpretation	H	Explain what findings mean for the problem

Answers: 1-C, 2-A, 3-D, 4-B, 5-E, 6-F, 7-G, 8-H.

15 Part O — Viva / Oral Defense Practice

Imagine your teacher asks: *"Defend your research problem in 60 seconds."*

Use the structure:



Your 60-second defense notes:
Problem: Gap: Objective: Evidence/data:

Problem: Fixed-time traffic signals cannot adapt to real-time traffic, so vehicles wait long and roads stay congested.

Gap: Existing work often uses complex setups; a clear, reproducible comparison of a simple DQN against fixed-time and actuated control at one junction, with measurable targets, is still needed.

Objective: Train a DQN controller in SUMO within 12 weeks to cut average vehicle waiting time by at least 15% compared with fixed-time control.

Evidence/data: SUMO logs of waiting time, queue length and throughput over 20 runs per controller, compared using mean +/- SD and a t-test.

16 Part P — Unit I Master Check

Can I do it without looking at the PPT?

Rate yourself honestly: **1 = not yet, 2 = partly, 3 = confidently**.

Skill	1	2	3
Explain what research is and why it matters			✓
Identify the key elements of research			✓
Generate sources of research problems			✓
Distinguish a topic from a research problem			✓
Evaluate a research problem using quality criteria			✓
Identify common selection errors			✓
Define scope and boundaries			✓
Write SMART objectives			✓
Choose an appropriate research approach			✓
Distinguish theoretical and empirical research			✓
Classify primary/secondary data			✓
Classify qualitative/quantitative data			✓
Select a suitable collection tool			✓
Explain analysis vs interpretation			✓
Select suitable research instrumentation		✓	
Explain why calibration and repeatability matter		✓	

My one-minute takeaway

The most important idea I learned in Unit I is:
A good research project starts with a clear, researchable and feasible problem, with defined scope and SMART objectives.

The concept I still need to revise is:
The difference between analysis and interpretation, and calibration/repeatability of instruments in more depth.

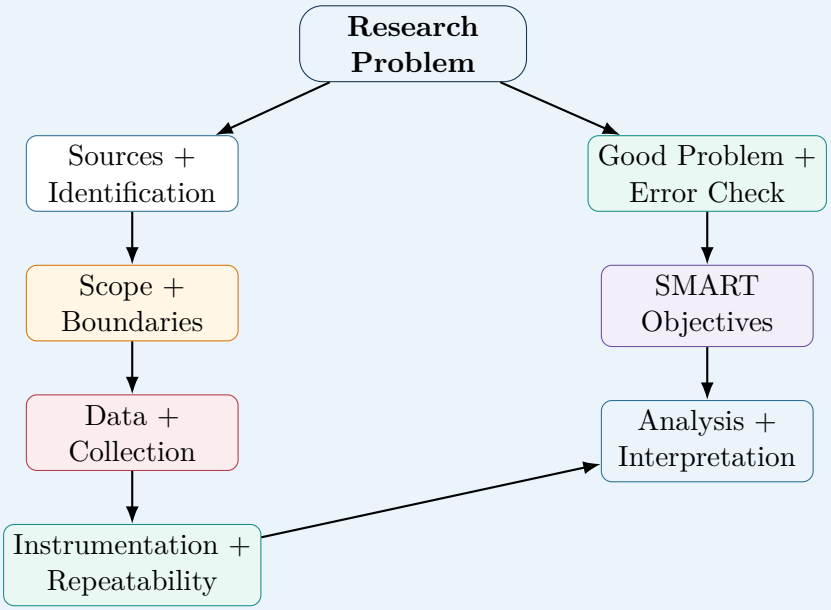
One research problem I can now formulate is:
How effectively can a DQN-based controller reduce average vehicle waiting time at a 4-way intersection compared with fixed-time signals?

Exit Ticket: Researcher Mindset

Complete the sentence:
"A good researcher does not start by asking what tool is trending? *; a good researcher first asks* what problem needs solving? *."*

Explain your answer in one or two lines:
Technology is only a tool. Research begins with a clear, meaningful problem and question; methods and tools are chosen afterwards to answer it.

Unit I in one visual



End of Unit I Worksheet — Introduction to Research Problem